Phase 1: Course and Lecture Preparation and Delivery Support

This phase focuses on helping a professor prepare for a course or an individual lecture. The assistant can consider course objectives, the topic to be covered, available class time and the expected student level. It can help decide what should be covered, in what sequence, and how much time should be spent on each part.

Possible Capabilities:

- Prepare a structured lecture plan
- Suggest the sequence of concepts
- Recommend important reading or reference material
- Suggest real-world and professional examples
- Plan classroom activities
- Create a rough time schedule for the lecture
- Suggest professional or industry applications
- Identify possible misconceptions students may have
- Provide additional explanations for difficult concepts
- Suggest interactive ways of presenting a topic

Technical Approach:

A language model can generate most of this content. Course documents such as the syllabus and previous material can be provided as context so that the suggestions remain relevant to the course. The professor should be able to review and edit the generated plan.

Phase 2: Learning Trails and Post-Class Continuity

This phase focuses on creating a trail around a topic so that students can be prepared before a class and can continue exploring after it.

Possible Capabilities:

- Generate pre-class teasers or introductory questions
- Create short post-class summaries
- Suggest further reading and reference material
- Generate practice activities/ questions
- Suggest connections with previously covered topics

Technical Approach:

This can largely be built using structured language model prompts and the course knowledge base. The assistant can use the lecture topic and generated plan as context for creating material before and after the class.

Phase 3: Assessment

This phase goes beyond simply generating a question paper. The idea is to create meaningful assessments and provide useful feedback about what a learner may need to improve.

Possible Capabilities:

- Generate quizzes and question papers
- Support different types of questions
- Generate questions at different difficulty levels
- Create questions for different levels of understanding
- Generate answer keys and model solutions
- Provide structured feedback based on expected answers ‘

Technical Approach:

Question generation can be handled well by language models when the topic, difficulty and format are clearly specified. Feedback can be based on rubrics and expected answers. The professor should remain responsible for reviewing important evaluation decisions.

Phase 4: Feedback Loop and Course Improvement

This phase uses feedback and assessment outcomes to help improve future teaching. The objective is to identify patterns that can help the professor understand which parts of a course may need improvement.

Possible Capabilities:

- Summarize recurring difficulties
- Identify topics that may need revision
- Analyze common mistakes
- Summarize class feedback
- Suggest improvements for future lectures

Technical Approach:

This would require structured data from assessments or feedback forms. The AI can then summarize patterns and suggest possible actions. Basic versions can work with sample or manually uploaded data.

Phase 5: Timeline and Course Management

This phase focuses on smaller teaching-related tasks and timelines. It can help the professor keep track of activities that need to happen throughout a course.

Possible Capabilities:

- Maintain a teaching timeline
- Track upcoming quizzes and assignments

- Suggest preparation tasks before important dates
- Remind the professor about pending activities

Technical Approach:

This can be implemented using an application logic, a calendar or task system, with AI used mainly to provide suggestions and organize tasks.

Proposed Phase to work on: Phase 1 to start with and then it can be extended to phase 2, once and if phase 1 is completed.

Proposed Technology Stack:

Frontend: ReactJS
Backend: FastAPI
Language Model: Gemini Flash
Course Knowledge: RAG-based document retrieval
Vector Database: ChromaDB
Application Database: PostgreSQL
AI Workflow Orchestrator: LangGraph (optional, can be a simple step-wise workflow too)